# Collecting Like Terms

## PRACTICE EXERCISE

Simplify:

1 $b + c + c + b$

2 $3t + t + u$

3 $\bigstar + \bigcirc + \bigcirc + 2\bigstar$

4 $\square + \blacksquare + (\blacksquare + \blacksquare) - \square$

5 $r + r + r + r + r =$

6 $10s - s - s - s =$

7 $17t + 4t + 2u - u + t =$

8 $a + b + c + a + b + c =$

9 $10v + 11v + 3w + 2w$

10 $6x + 0.5x + y + 2.5y$

11 $\frac{3}{7}z + 2z + 4a + \frac{5}{4}a$

12 $ab + bc + ab + 3bc$

13 $2v + w + 3w - 2v =$

14 $10x + 11x + 12y + 13y + 14x =$

15 $0.1i + 0.2j + 0.3i + 0.9j =$

16 $kl + 2l + 7l + 3k + lk$

17 $-5d - 3e + 7e - 6d$

18 $f(g) + 2gf - fg + fgh$

19 $\varphi + 3 + \varphi + \varphi + 1$

20 $62q + 108q + 4q - 2q^2$

21 $-2m - n - 4m$

22 $p + q + r + pq + qp + pr$

23 $\frac{3}{2}t + \frac{1}{3}u + \frac{2}{6}t - \frac{4}{5}u$

24 $(3 + 4)a - (-2a + 4a)$

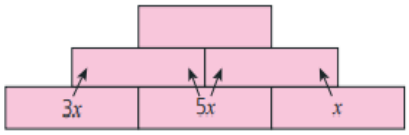
## ACTIVITY: Race to the top

**ATL**

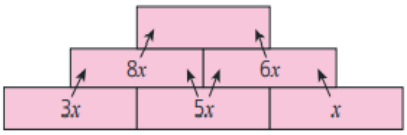
- Transfer skills: Apply skills and knowledge in unfamiliar situations

### Collecting like terms in a pyramid

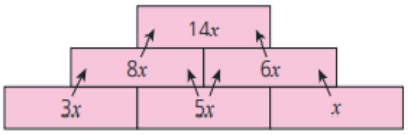
Step 1: Start at the bottom row and **add** adjoining (touching) boxes together to give the box directly above them.



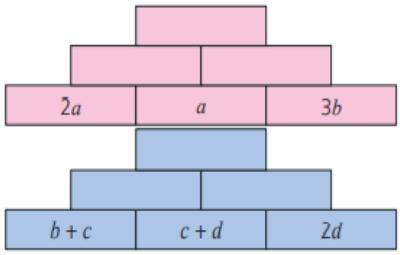

Step 2: Repeat for the next row.

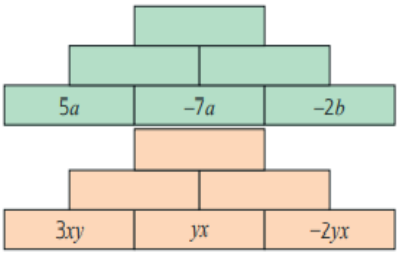



Step 3: Get the final answer at the top of the pyramid.

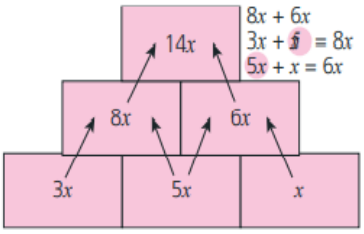



Try these pyramids:




### EXTENSION




**Assessment opportunities**

## PRACTICE EXERCISE

Simplify:

1 $a + 3 + a + 2 =$

2 $3b + 4 + 2b + 10 =$

3 $c + 5 + 2c + 1 =$

4 $d + e + 4 + d + e =$

5 $4f + 3 - 2f + 2 =$

6 $9g - 2 + g + 11 - 10g =$

7 $5h + 5i + 10h + 11i =$

8 $4.5j + 4.5 + 2j - 2.3 =$

9 $-k + 100 + 42 + 5k - 230 =$

Question 3: Simplify the following expressions

(a) $3a + 2b + 4a + b$ (b) $7y + 5y + 2h + 2h$ (c) $g + 8a + 2a + g$

(d) $7m + 7p + 8m + p + 2p$ (e) $9e + 2 + e + 2$ (f) $4 + 3a + 2a + 8$

(g) $2y + 4 + 3y - 1$ (h) $8 + 3w - w - 3$ (i) $5 - 4s - 2 + 10s$

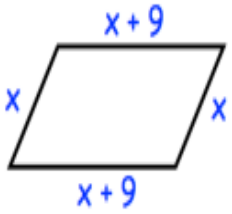
(j) $3x + 6y + 5x - 2y$ (k) $6m - 2s + 11s + m$ (l) $2a + 3b - 2 + a + 3b + 4$

(m) $3a - 2b + a - 5b$ (n) $2x - 2y - 6x + 5y$ (o) $y - 4m - 3y - 5m$

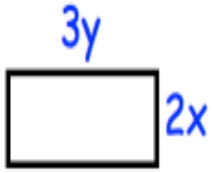
(p) $7p - 2q - q + 3r + 4r$ (q) $11c + 8d - 6c - 11d$

Question 1: Write down the perimeter of each shape below

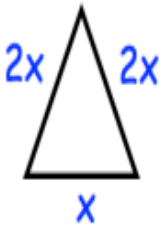
(a)




(b)



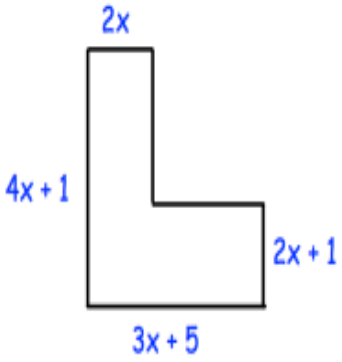

(c)




Question 2: A square has a side length of $3x$.
Find an expression for the perimeter of the square.

Question 3: $6x + 7y + x - 8y = 7x - y$
Write down three other expressions that are equal to $7x - y$

Question 4: Find an expression for the perimeter of this shape




Answers